KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

MID-TERM REPORT ON

A CASE BASED INVESTIGATION OF BATCH TYPE PYROLYSIS REACTOR COMMUNITY BASED GREEN SOLUTION

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LIST OF ABBREVIATION

ABS	Acrylonitrile Butadiene Styrene
CAD	Computer-Aided Design
FCC	Fluid Catalyst Cracking
FTIR	Fourier-transform infrared spectroscopy
GCMS	Gas Chromatography and Mass Spectrometry
HDPE	High-Density Polyethene
LDPE	Low-Density Polyethene
LPG	Liquid Petroleum Gas
MCM	Mobil Composition of Matter
NTIC	Nepal Technology Innovative Center
PA	Polyamide
PET	Polyethene Terephthalate
PP	Polypropylene
PPS	Polyphenylene Sulfide
PS	Polystyrene
PVC	Polyvinyl Chloride
ZSM	Zeolite Socony Mobil

ABSTRACT

The global problem of plastic waste has serious environmental impacts and lacks a long-lasting solution. In order to minimize and reuse plastic, plastic pyrolysis is introduced which converts plastic waste into useful products through heat and offers a promising solution. This study focuses on using pyrolysis to manage low-density polyethylene (LDPE) plastic waste through a thermal decomposition process.

A pyrolysis unit was designed and built to process LDPE. The pyrolysis was conducted in the absence of oxygen at a temperature of 450°C and a pressure of 2 bar, resulting in the production of pyrolysis oil. The produced pyrolysis oil was tested and showed a high energy value of 37 MJ/kg through a bomb calorimetry test. FTIR analysis found that the oil's chemical bonds are similar to those in diesel, with matching bending and stretching bonds in the wavenumber range of 2700 to 3000 cm^{-1}. GC-MS analysis revealed that the oil contains carbon numbers from C9 to C26, similar to the C10 to C30 range in diesel. Further analysis showed that the pyrolysis oil has about 83% carbon and 16.75% hydrogen, compared to diesel's 87% carbon and 13% hydrogen. The sulfur content in LDPE is 0.25%, whereas diesel contains only 0.0015%. These results suggest that pyrolysis oil is chemically similar to diesel, making it a viable alternative fuel.

This study shows that plastic pyrolysis can effectively manage plastic waste, providing environmental, and social benefits.

CHAPTER 1. INTRODUCTION

1.1.Background

Plastic is a synthetic or semisynthetic polymer made up of natural materials such as cellulose, coal, natural gas, salt, and Pyrolysis oil through a polymerization or polycondensation process. Being a revolutionary product, it is significantly used in every sector from industries to everyday life in huge quantities. The production rate of plastic is increasing consequentially each year across the globe and is expected to reach 445 metric tons in 2025. [1]

Figure 1 shows the annual production of plastic waste from 1950 to 2022 [1]

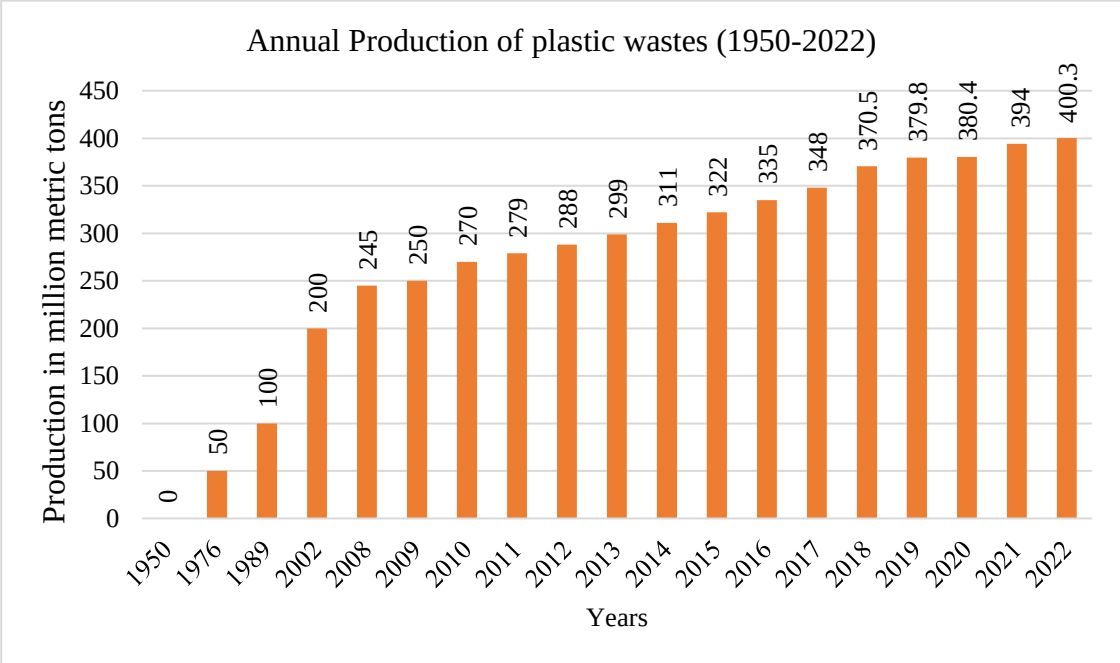

Figure 1: Annual Production of Plastic Waste from 1950 to 2020

Plastic waste has become a significant problem in recent decades due to the rising consumption of this versatile material. Global plastics production has doubled since the start of the century, reaching almost 400.3 million metric tons per year in 2022. [2] Although plastic products typically last around 10 years, they can take up to 500 years to decompose, depending on their composition and disposal methods. The world is struggling to manage the increasing production of disposable plastic products, leading to severe plastic pollution. This issue is particularly noticeable in underdeveloped Asian

and African countries, where waste collection systems are often ineffective, nonexistent. The accumulation of plastic waste poses serious threats to marine life, ecosystems, and human health LDPE, commonly used in plastic bags, and packaging, is a major contributor to plastic waste due to its widespread use and low recycling rates. [2]

Plastic waste can be broken down into usable products such as fuel oil, char, and gases through a process called plastic pyrolysis. This process occurs without oxygen and has several benefits, including the ability to handle contaminated and mixed plastics, reduce landfill use, and produce useful products that can serve as alternative energy sources. About 75–80% of the liquid oil produced from LDPE pyrolysis can replace traditional fuels.

1.2.Problem Statement

In the village of Bethanchwok, located in the Kavre district, residents engage in small-scale khuwa production as a primary business activity. This production relies on khuwa-making machines powered by fossil fuels such as diesel or petrol. However, there is an

Figure 2: Khuwa Processing Machine

opportunity to replace these conventional fuels with pyrolyzed fuel, which can be produced at a lower cost compared to diesel and petrol. Moreover, this pyrolyzed fuel can be derived from locally available LDPE plastic waste, providing a sustainable solution to plastic waste management issues in the community. This project investigates

the use of a batch-type pyrolysis reactor as a community-based green solution to manage plastic waste and provide a sustainable fuel alternative.

1.3.Objective

1.3.1. Primary Objective

- To re-model the steering-type batch reactor.
- To observe and analyze the characteristics of pyrolyzed oil.

1.3.2. Secondary Objective

- To study the socio-economic feasibility of the plant in the sites.
- To test and analyze the product

1.4.Scope and limitation

- Limited to LDPE: Cannot process PET & PVC, reducing versatility.
- Small Batch Size: Processes only 1.5 kg per batch, limiting throughput.
- Combustion-Only Oil: Pyrolysis oil is suitable only for combustion, limiting its value.

CHAPTER 2. METHODOLOGY

2.1.Literature Review

Pyrolysis can be accomplished in both thermal and catalytic processes. However, the catalytic process provides a lower operating temperature with a greater yield of liquid oil for most plastics with the right catalyst selection. Similarly, there are various other factors that affect the rate and yield of the product. Some major factors are temperature, pressure, type of plastic selected, absence of oxygen, etc.

Table 1: Different Types of Thermoplastics and Their Properties [3]

Type of plastics	Moisture (wt.%)	Fixed carbon (wt.%)	Volatile (wt.%)	Ash (wt.%)
PET	0.46	7.77	91.75	0.02
	0.61	13.17	86.83	0.00
High-density polyethylene	0.00	0.01	99.81	0.18
	0.00	0.03	98.57	1.40
PVC	0.80	6.30	93.70	0.00
	0.74	5.19	94.82	0.00
Low-density polyethylene	0.30	0.00	99.70	0.00
	–	–	99.60	0.40
Polypropylene	0.15	1.22	95.08	3.55
	0.18	0.16	97.85	1.99
Polystyrene	0.25	0.12	99.63	0.00
	0.30	0.20	99.50	0.00
PE	0.10	0.04	98.87	0.99
ABS	0.00	1.12	97.88	1.01
PA	0.00	0.69	99.78	0.00
Polybutylene	0.16	2.88	97.12	0.00

The table mentioned shows the plastics and their properties. Plastics like PET, HDP, and PVC have lower volatility than LDP which has volatility around 99.70%. Also, carbon contain in LDP is also very low which results in low formation of char (not desired product). LDPE contains low percentage of moisture than other plastics which makes this project run more efficiently and in less period of time.

Table 2: Types of Pyrolysis and Their Yield Percentage [3]

Pyrolysis type	**Pyrolysis products (%)**			**Liquid oil (composition %)**
	Liquid oil	**Gases**	**Char**	
Thermal Pyrolysis	80.8	13	6.2	Styrene
Catalytic Pyrolysis (Natural Zeolite)	54	12.8	32.8	Styrene (60.8) Methyl styrene (10.7) Azulene (4.8) 1H-Indene (2.5) Ethylbenzene (1.3)
Catalytic Pyrolysis	50	22.6	27.4	Styrene (15.8) Benzene (16.3) Propenyl benzene (4.2) Propyl benzene (3.5) Isopropyl benzene (8.1) Alpha-Methyl styrene(38.4)

The table mention shows the different types of pyrolysis. Pyrolysis is the thermal degradation of plastic waste at different temperatures (300^0 C–700^0C) in the absence of oxygen, to produce liquid oil. Thermal pyrolysis does not use any type of catalyst. With the thermal cracking the heat supplied will crack the molecule of polymer and

changes it into liquid fuel. Also, the yield percentage of this method is high which is above 80 %. So thermal cracking is applied in this project

2.1.1. Proximate Analysis of different types of plastic and Diesel

Table 3: Proximate Analysis of Different Types of Plastic and Diesel [4]

Properties	ASTM Standards	PET	HDPE	LDPE	PVC	PP	PS	Mixtures PE+PP+PS	Diesel
Calorific Value (KJ/KG)	ASTM D240-19B	28	40-49.9	37.4-46.6	18-210	40-46.4	37.7-43	41-43	43-44
API gravity	ASTM D287-20	-	28	48	39	33	-	38.98	38
Viscosity at 40C(mm^2)	ASTM D445-21	-	5.08	5.55	6.36	4.09	1.4	2.52	1.9-4.1
Density at 15C(g/cm^2)	ASTM D1298-12e1	0.9	0.89	0.78	0.85	0.86	0.85	0.83	0.8-0.83
Ash Content (wt. %)	ASTM D482-13	-	0	0.02	-	0	0.006	1.04	0.01
Octane Number	ASTM D2700-17	-	85	-	-	87	-	-	15-25
Cetane Number	ASTM D613-21	-	95	85	-	97	90-98	60	52
Pour Point	ASTM D97-20b	-	-5	-7	-15	-9	-67	-42	6
Flash Point	ASTM D93-21	-	48	41	40	30	26	30	52

The table above shows a proximate analysis of different types of plastics. Among these plastics we choose LDPE. LDPE has good quality of yield when pyrolysis and the properties are near to the properties of diesel.

2.1.2. Ultimate Analysis of LDPE

Table 4: Ultimate Analysis of LDPE and Diesel [5]

S.No.	Element Contains	LDPE Weight (%)	Diesel Weight (%)
01	Carbon (C)	83.00	87
02	Hydrogen (H)	16.75	13
03	Nitrogen (N)	0.00	No specific Nitrogen
04	Sulphur (S)	0.25	0.0015

The table mentioned the Ultimate analysis of LDPE and Diesel by weight (%). The Carbon and Hydrogen numbers are nearly equal among these two components. Where whereas Sulphur contains is bit more in LDPE.

2.2.Theoretical Framework

LDPE plastic pyrolysis involves heating plastic waste in the absence of oxygen to break it down into smaller molecules like oils, gases, and solid residue. These products can be used for making fuels or chemicals. LDPE pyrolysis helps manage plastic waste that's hard to recycle conventionally, though it requires careful control of process parameters and consideration of environmental impacts. Overall, it's a promising way to reduce plastic pollution and recover useful materials from waste

The concept, design, and ideas for the Plastic Pyrolysis are mentioned below:

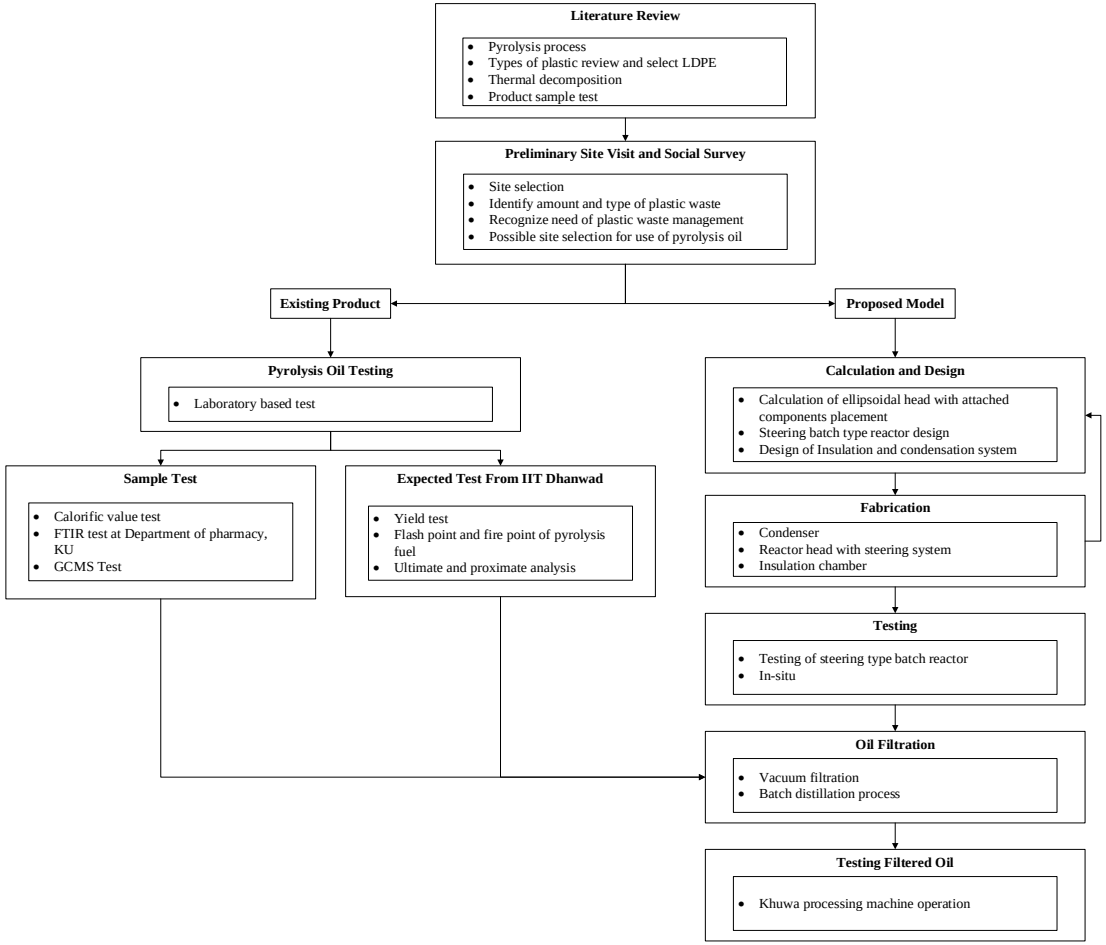

Figure 3: Proposed Methodology

The above proposed methodology explains as:

The proposed plan starts with studying the pyrolysis process, focusing on LDPE (a type of plastic), how it breaks down with heat, and testing the resulting products. Then, a visit to potential sites will be made to choose the best location, check the amount and type of plastic waste, and see the need for managing plastic waste. The design phase involves planning the reactor and its parts, including the fabrication of the reactor and condenser. Lab tests will be done for the existing product pyrolysis oil, and further tests at IIT Dhanwad will measure its yield, flash point, and fire point, and analyze its composition. This test will determine the oil's energy content and its chemical makeup. The next step is proposing our model for the modification of existing products and building the necessary equipment like the condenser and reactor. The system will then be tested as in-situ parameters and further oil filtration and distillation will be done using a batch distillation process and vacuum filtration technique. Finally, the filtered oil will be tested in a Khuwa processing machine.

2.2.1. Thermal Cracking

The primary focus of this project is to utilize thermal cracking to convert waste plastic into valuable petroleum products. This is achieved through the process of pyrolysis, which involves the thermal decomposition of plastics at high temperatures in an inert atmosphere. During pyrolysis, the chemical structure of the plastics breaks down into hydrocarbon compounds. Given that plastics are derived from petroleum, the resulting oil from this process has a high caloric value, making it a viable alternative fuel. The efficiency of the pyrolysis process improves with an increase in temperature, thereby accelerating the rate of thermal decomposition. This approach not only provides an effective disposal method for waste plastic but also generates high-energy fuel, contributing to sustainable waste management and energy production. [7]

2.2.2. Study Design

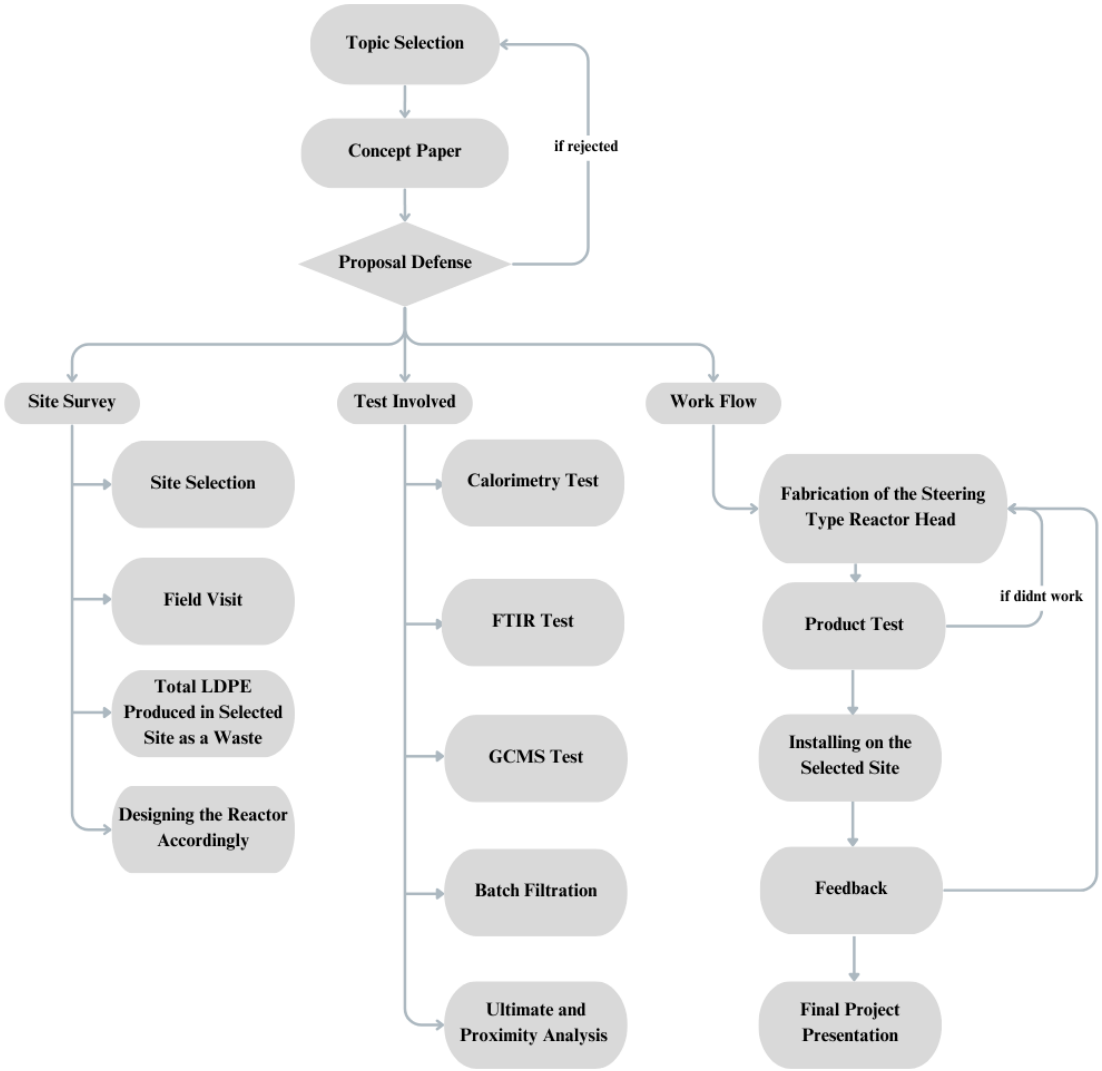

Figure 4: Flow Chart of Project

The above flow chart of the project is described below:

The proposed methodology begins with the selection of a research topic, followed by writing a concept paper on the chosen topic. This leads to the proposal defense stage, where the proposal is presented and defended. If the proposal is rejected, the process

starts with topic selection. If accepted, the work is divided into three phases. The first phase, the site survey, involves site selection, conducting a field visit, assessing the total amount of LDPE plastic produced at the site, and selecting the right domain for replacing traditional fuel with pyrolysis oil. The second phase, the test includes performing calorimetry tests to measure the energy content of the pyrolysis oil, conducting FTIR and GCMS tests, and characterizing the oil. The third and final phase, the workflow, involves modeling the steering-type reactor, testing its production capabilities, installing the reactor at the selected site, collecting feedback from the installation and operation, and presenting the final project findings and results.

2.3. Cad Design and Components of Pyrolysis Unit

2.3.1. Isometric View of Fabricated Product

The Figure below shows the Isometric view of the fabricated product. It is designed in Solid Works Software. The reactor is placed in the center top of the Reactor holder. On top of the reactor, there is the ellipsoidal head that has a pressure gauge, safety valve connected, and a agitator is also connected to the motor. The condenser is placed at the right top side of the reactor holder which is connected to the reactor head through a copper hosepipe.

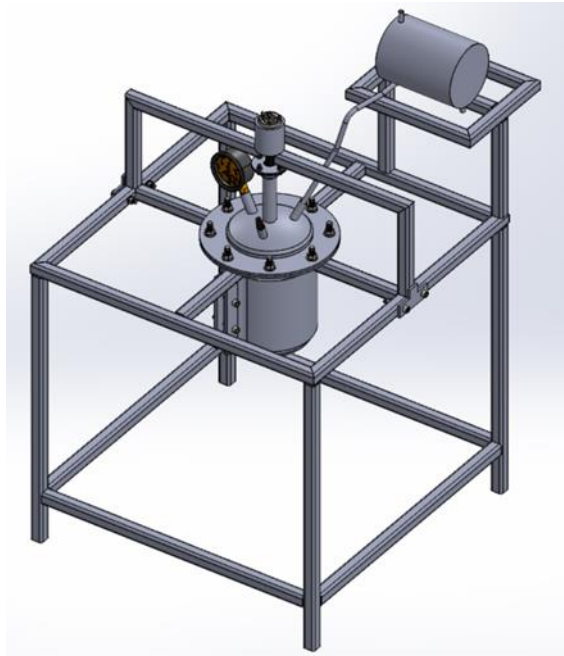

Figure 5: Isometric View of Reactor

The list of parts that are seen in the above drawing is further explained below

2.3.2. Reactor

A reactor is a combustion chamber made up of stainless steel where the dry fine plastic granule is fed inside an air-tight chamber and heated with a mica band heater.

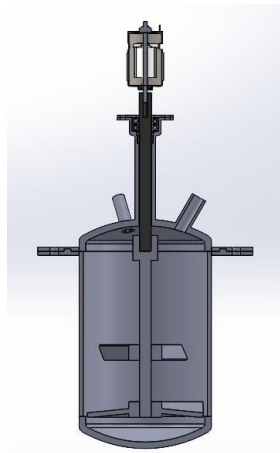

Figure 6: Cross-Sectional View of Reactor

It consists of safety valve, pressure gauge, and steam outlet pipe and agitator. The maximum temperature limit for the Reactor is 600 ^{0}C. [7]

2.3.3. Condenser

The condenser cools hot gases from the reactor to turn them into liquid fuel. This condenser is a helical shell and tube heat exchanger, where a copper coil is placed inside a cylindrical mild steel shell. The hot gases flow through the copper coil, which is surrounded by continuously flowing cold water. As the hot gases pass through the coil, they transfer heat to the cold water, which then warms up and exits the shell. This cooling process causes the hot gases to condense into liquid fuel, which is collected in a fuel collection chamber. The copper coil's excellent heat transfer ability and the durable mild steel shell ensure efficient cooling and condensation.

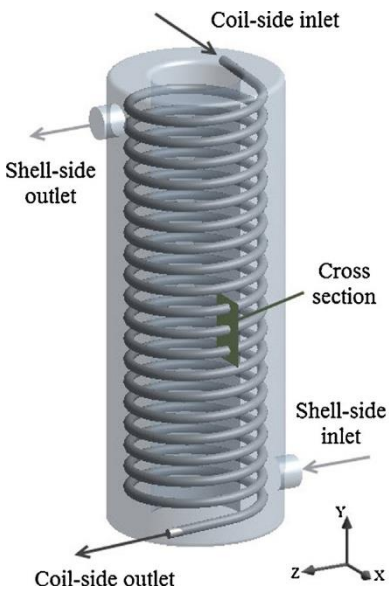

Figure 8: Cross Sectional View of Condenser

Figure 7: Fabricated Condenser

2.3.4. Pyrolysis Oil Collector

The pyrolysis oil collector is a container where the cooled liquid fuel is collected after the hot gases from the reactor are cooled down and turned into liquid through the condensation process.

2.4.Governing Equations

2.4.1. Conduction

Conduction is the process by which heat energy is transmitted through collisions between neighboring atoms or molecules. The temperature of the reactor ranges from 400^0 C to 500^0 C. [3]

$q = -kA\nabla T$ ……1 [8]

Where,

q=Local heat flux density(W/m^2)

k=Material's conductivity (W/ (m. k))

∇T= Temperature gradient (°C/m)

A= Exposed surface area ($m^{2)}$

2.4.2. Convection

Convection is the process by which heat is transferred from a solid surface to a nonsolid, such as air or water. It is applied when the plastic changes its phase to vapor.

$Q = hA\nabla T$……2 [9]

Q = Heat transfer rate (J/s)

h= Convection heat-transfer coefficient ($W/(m^2k)$)

A= Exposed surface area (m^2)

△T = Temperature difference Radiation ((°C/m)

The rate of heat transfer by emitted radiation is determined by the Stefan-Boltzmann law of radiation.

$Q = \sigma eAT^4$……3 [10]

$\sigma = 5.67 \times 10^{-8}\ W/m^2.\ K^4$

Where,

σ = The Stefan-Boltzmann constant ($W/m^2.\ K^4$)

e = emissivity

A = the surface area of the object (m^2)

T = absolute temperature in kelvin

2.5.Modification On Existing Product

Different types of modification are done on previously fabricated reactors which are listed below:

2.5.1. Ellipsoidal Head

The reactor head has an ellipsoidal shape and includes a pressure gauge, a safety valve, a gas outlet port, and a new steering system. The steering system is powered by a DC motor placed 10 cm away from the head. This motor drives an agitator that passes through a leak-proof structure shaft equipped with components like a brass bush, bearing, O-ring, and snap ring. The top of the agitator has fin-type blades that stir the semi-liquid plastic inside the reactor. This stirring action helps to evenly distribute heat throughout the plastic, ensuring uniform processing.

2.5.2. Steering System

The steering system modification includes an agitator that stirs the molten LDPE (Low-Density Polyethylene) inside the reactor. This stirring helps spread heat evenly throughout the system. The agitator is powered by a shaft connected to a DC motor, and it uses fin-shaped materials inside the reactor to improve heat distribution. To prevent gas from escaping through gaps, the steering system is designed with precise seals that ensure everything stays tightly closed and gas-tight.

2.5.3. Condensation Process

The condensation process is improved by adding a copper coil to the system. Water flows continuously through this copper coil, which helps to cool down and condense the hot gases more effectively. This setup ensures that heat is removed efficiently, making the condensation process work better.

2.5.4. Reactor Insulation Addition

To further enhance the system's efficiency, insulation is added to the reactor. This insulation helps to retain heat within the reactor, reducing energy loss and ensuring a more consistent temperature throughout the process. By maintaining a stable temperature, the reaction conditions are kept optimal, which improves the overall performance and efficiency of the system. The insulation also contributes to safety

by preventing the external surface of the reactor from becoming too hot, thereby reducing the risk of burns or other heat-related hazards.

2.6.Test Setup for Characterization of Pyrolysis Oil

2.6.1. Bomb Calorimetry Test

This measurement confirms the oil's efficiency as a fuel by quantifying the amount of energy it can produce when fully burned. With this high calorific value, the pyrolyzed oil demonstrates its potential for effective use in heating and power generation.

Figure 9: Calorimetry Testing Machine

Figure 10: Bomb Calorimeter

2.6.2. FTIR of Plastic Pyrolysis Oil

FTIR is an analytical technique used to identify organic, polymeric, and, in some cases, inorganic materials. The method works by measuring how different chemical bonds absorb infrared light at varying wavelengths. In case of pyrolysis oil, a sample is

Figure 11: FTIR Testing at Pharmacy Lab KU

exposed to an infrared light beam through Attenuated Total Reflection (ATR) method as the light passes through or reflects off the sample, specific wavelengths are absorbed by the material's molecular bonds. This absorption pattern is then recorded as an interferogram. A mathematical algorithm called Fourier transform is applied to convert this interferogram into an infrared spectrum, displaying the absorbed light intensity as a function of wavelength. [11]

2.6.3. GCMS Test

GCMS plays a crucial role in analyzing pyrolysis fuel, helping to determine its chemical composition and quality. Through GCMS testing, various chemical compounds present in the fuel are identified and measured, including hydrocarbons (such as alkanes and alkenes), oxygenated compounds (like alcohols and acids), nitrogenous compounds (such as amines), sulfur compounds (like sulfides), and other complex molecules like phenols. This detailed analysis is essential for ensuring the fuel's quality and suitability for use.

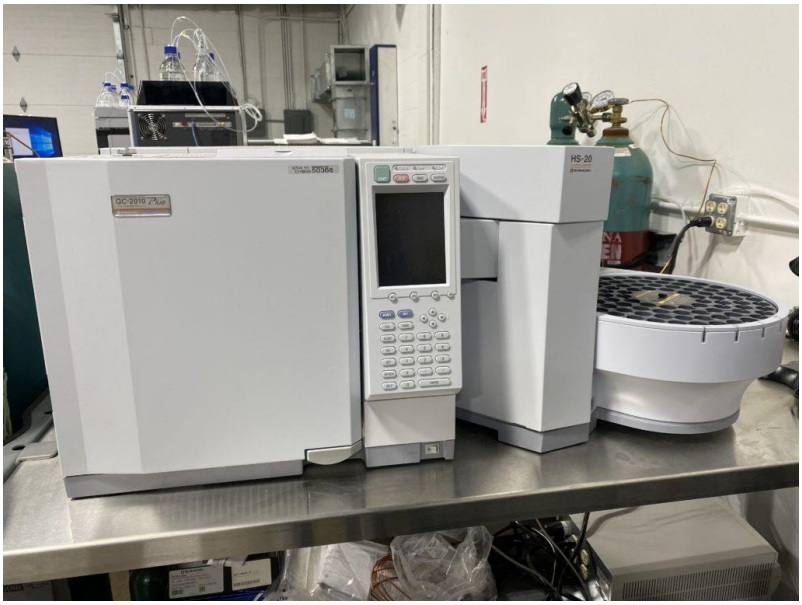

Figure 12: GCMS Testing Setup

Additionally, GCMS testing can detect toxic components, including certain sulfur and nitrogen compounds. Identifying these harmful substances is vital, as they can then be filtered out using appropriate solutions, resulting in a cleaner and safer fuel for community use. This process not only enhances fuel quality but also aligns with the goals of promoting an environmentally friendly and sustainable green solution for the community. [4]

2.7.Calculation

Calculation of Torque required to Steer Plastic inside reactor.

Dynamic viscosity of LDPE at 200 °C and 786 s^{-1}

Rate of shear stress = 1.7×10^{-5} Nmm^{-1} [12]

Radius of Reactor (R) = 150mm

Clearance between wall (h) = 40mm

Speed of Shaft (N) = 50rpm

Length of Reactor (L) = 200mm

$$\text{Force(F)} = \frac{4\pi^2\mu R^2 LN}{60h} \quad \text{.....} (4)$$

$$F = \frac{4\times\pi^2\times1.7\times10^{-5}\times150^2\times200\times50}{60\times40} \quad \text{.....} (5)$$

F = 62.91N

$$\text{Torque (T)} = F \times R \quad \text{.....} (6)$$

T = 62.91×150

T = 9437.80Nmm^{-1}

So, the motor required will overcome this torque.

CHAPTER 3. WORK ACCOMPLISHED

3.1. Bomb Calorimetry Test of Pyrolysis Oil

The test was carried out in NTIC at Kathmandu University, Dhulikhel. A sample of 1 gram of pyrolysis oil was taken and poured into to bomb calorimeter. It is then pressurized to 20 bar and ionized using two electrodes of nichrome wire of size 0.1 mm to 0.3mm. On the other hand, a device gradually measured an increase in heat energy to complete the combustion of the sample and the heat energy that was produced was **37MJ/Kg** was obtained which is the calorific value of plastic pyrolysis oil.

3.2. FTIR of Plastic Pyrolysis Oil.

The test was carried out in the Department of Pharmacy of Kathmandu University, Dhulikhel. A sample was directly placed in an ATR crystal and direct measurement and evaluation were started. The figure below shows the result of an FT-IR investigation of pyrolysis oil. This test used the principle of the Attenuated Total Reflection (ATR) method to observe the sample. The plastic oil samples had peaks in the range of 399 cm^{-1} and 4700 cm^{-1} of wavenumber. Additionally, the presence of C-H stretches for the methyl and methylene groups can be found in between the wavenumber 1800 cm^{-1} and 3200 cm^{-1}. In between 1600 and 1650 the existence of C=O starch for ester, carboxylic acid, ketones, and aldehyde group is indicated by two peaks.

Table 5: Types of Vibration and Wavenumber of FTIR Test of Pyrolysis Oil

S.N.	Type of Vibration	Wavenumber cm^{-1}
1	Skeletal C-C vibration -CH_3 frequencies	692-736
2	Skeletal C-C vibration -CH_3 frequencies	827-887
3	Methyl C-H asymmetrical Bend	1462-1479
4	Methyl C-H symmetrical Stretch	2829-2873
5	Methyl C-H asymmetrical Stretch	2916-3007

The table above mention shows different types of vibration with their specific wavenumber. It also shows maximum bending and stretching wave number in vibration.

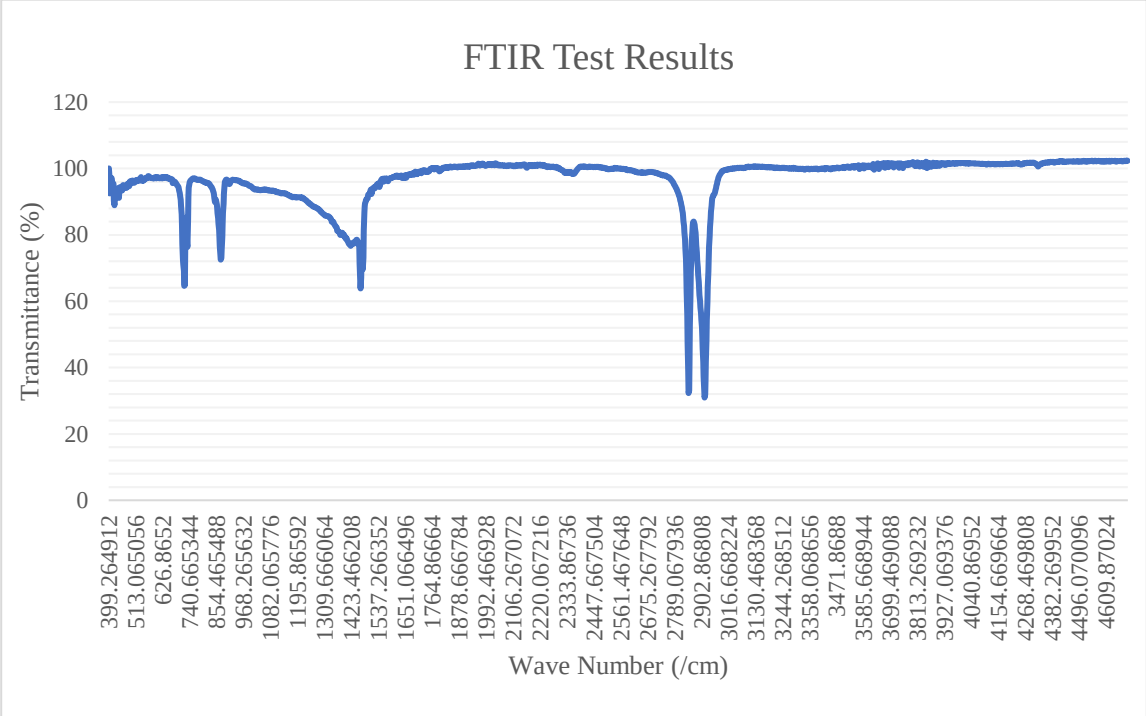

Figure 13: FTIR Spectrum of Pyrolysis Oil

3.2.1. FTIR of Diesel.

The infrared spectrum of diesel lacks carbonyl groups and does not show any absorption in the fingerprint region, indicating no presence of C–O bonds. This absence accounts for the reduced peak intensity of around 1740 cm^{-1} as the diesel concentration in a mixture increases. Consequently, the intensity of the carbonyl group peak can be utilized to determine the biodiesel concentration in such mixtures. [13] The type of vibration and stretches and bending of the C-H atoms on diesel is as same as in pyrolysis oil. Symmetric stretches and asymmetric stretches are found on striking IR wavelengths of 2700 to 3000 cm^{-1} and asymmetric bending is found on wavelength 1200 to 1500 cm^{-1} also Skeletal C-C vibration -CH_3 frequencies are found at 600 to 900 cm^{-1}. [4]

Table 6: Types of Vibration and Wavenumber of FTIR Test of Diesel

S.N.	Type of Vibration	Wavenumber cm^{-1}
1	Skeletal C-C vibration -CH_3 frequencies	722.43
2	Skeletal C-C vibration -CH_3 frequencies	805.42
3	Methyl C-H asymmetrical Bend	1377.16
4	Methyl C-H symmetrical Stretch	2854.05-2922.60
5	Methyl C-H asymmetrical Stretch	2922.60-2954.30

The table above mention shows different types of vibration with their specific wavenumber. It also shows maximum bending and stretching wave number in vibration.

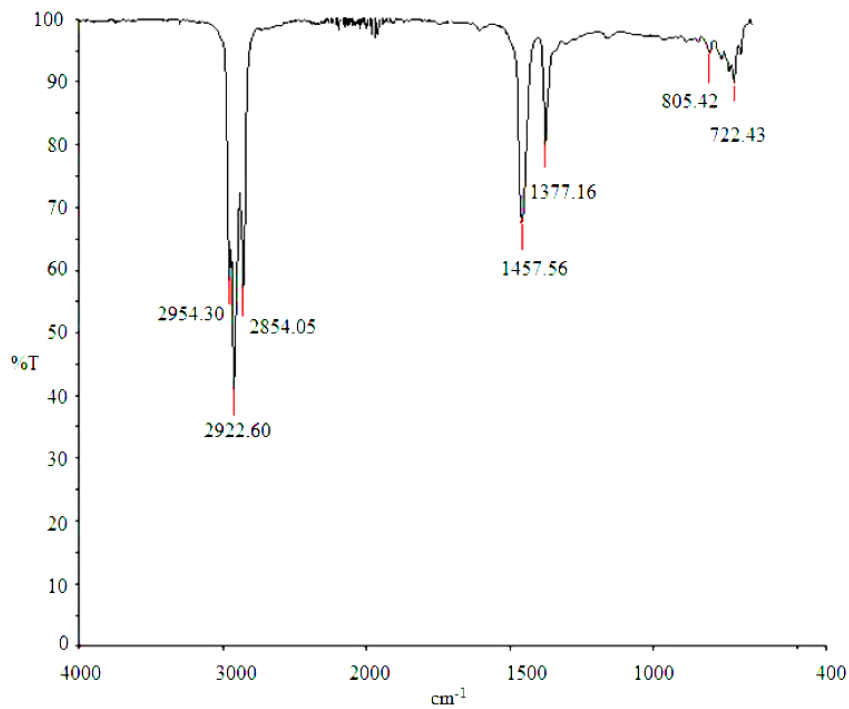

Figure 14: FTIR Test Result of Diesel.

3.3.Comparison Between Pyrolysis Oil and Diesel

Pyrolysis oil from LDPE plastic has some important differences from standard Diesel. It is thicker (with a viscosity of 5.55 mm^2/s compared to diesel's 1.9 to 4.1 mm^2/s),

which can cause problems with fuel injection and burning properly in engines. It is slightly less dense (0.78 g/cm³ versus diesel's 0.80 to 0.83 g/cm³), meaning it has a bit less energy per volume, possibly reducing fuel efficiency and requiring engine adjustments. The oil ignites more easily because its flash point is lower (41°C versus diesel's 52°C) and it catches fire at a lower temperature (30°C versus diesel's 52°C), making it more efficient than diesel. Chemically, pyrolysis oil has a bit less carbon (83% versus diesel's 87%) and more hydrogen (16.75% versus diesel's 13%), which might help it burn cleaner with fewer carbon deposits but produce more water vapor in the exhaust. It also has a narrower range of carbon atoms (C9 to C26 compared to diesel's C10 to C30), which might make its burning properties more consistent but affect how it vaporizes.

CHAPTER 4. BUDGET ESTIMATION AND GANTT CHART

4.1.Budget Estimation

Table 7: Budget Estimation

S.N.	Name of the Material	Quantity	Price (Rs)	Remarks
01	Mild Steel Sheet	1 Piece	5,000	TTC
02	Copper Hosepipe	3 Meter	2,000	Market
03	Gascade Sheet	2×2 ft^2	2,700	Market
04	Fasteners	20 Pair	400	Nut and Bolt
05	Control Valve	1 Piece	750	Market
06	Battery	1 Piece	500	Market
07	Pump	1 Piece	350	Market
08	Bearing	2 Piece	400	Market
09	Brass Bush	1 Piece	100	Market
10	Gear Motor	1 Piece	1,200	Market
11	Mild Steel Shaft	1 Meter	1,200	TTC
12	Costing for Rolling, Gas cutting, Gas Welding, and Brazing.		2,500	Workshop
13	Mild Steel Strip	1.5 Meter	300	TTC
14	Square Pipe	20 Feet	2,000	TTC
15	Safety Valve	1 Piece	1,200	Market
16	Pressure Gauge	1 Piece	1,400	Market
17	Beaker	1 Piece	450	Market
18	Oil Filtration Setup	1 Set	2,000	Market
19	Tap	3 Piece	2,400	Market
20	Glass Wool	3 Meter	700	Market
TOTAL			NRS 26,850	

The estimated total budget of the project is Rs. 26,850.

4.2.Gantt Chart

Weeks	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Topic Selection														
Concept Paper														
Proposal Defense														
Literature Review														
Field Visit														
Characterization of Pyrolysis oil														
Redesigning of Existing project														
Mid- Term Presentation														
Fabrication of Steering System Reactor														
Testing of the product														
Installation of product in selected sites														
Feedback and Analysis														
Report submission and presentation														
													Remaining Works	
													Completed Works	

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